

BERN ZOLLIKOFEN, Switzerland

WMO: 066310

Lat: 46.9908N

Lon: 7.4639E

Elev: 567

StdP: 94.70

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -7.3	(d) -12.4	(e) 1.4	(f) -7.9	(g) -9.8	(h) 1.7	(i) -6.0	(j) 9.3	(k) 8.9	(l) 7.8	(m) 6.4	(n) 2.0	(o) 20	(p) 0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	30.1	19.5	28.3	19.1	26.5	18.4	20.5	28.0	19.7	26.6	18.9	25.3	2.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.8	23.2	17.2	13.2	22.4	16.6	12.7	21.6	61.4	28.3	58.6	26.7	56.0	25.5	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	5.7	4.8	-12.4	32.9	2.7	1.6	-14.3	34.1	-15.9	35.1	-17.4	36.0	-19.3	37.2
			-12.7	22.2	2.8	1.0	-14.7	22.9	-16.3	23.4	-17.9	24.0	-19.9	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	0.5	1.2	5.3	9.6	13.2	17.6	19.3	18.7	14.4	10.3	4.8	1.1		
	DBStd	7.66	4.13	4.22	3.90	3.62	3.54	3.59	3.11	3.23	3.24	3.65	3.55	3.74		
	HDD10.0	1237	296	248	149	51	11	1	0	0	4	43	158	277		
	HDD18.3	3293	554	481	403	262	162	56	27	36	122	249	406	535		
	CDD10.0	1130	1	0	4	38	111	228	287	269	137	52	3	0		
	CDD18.3	144	0	0	0	0	3	34	56	46	4	0	0	0		
	CDH23.3	1332	0	0	0	4	49	333	510	398	37	0	0	0		
	CDH26.7	364	0	0	0	0	6	87	151	119	2	0	0	0		
(14) Wind	WSAvg	1.8	1.8	1.9	2.1	2.0	1.9	1.9	1.7	1.6	1.6	1.5	1.5	1.7		
(15) Precipitation	PrecAvg	1169	64	63	71	89	127	130	134	123	102	99	85	82		
	PrecMax	1471	156	148	262	197	225	269	292	216	176	174	181	162		
	PrecMin	905	18	13	27	19	47	53	47	17	51	16	5	5		
	PrecStd	160	39	37	47	43	50	50	52	48	36	38	48	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	13.6	18.7	23.9	27.2	31.4	32.6	32.7	26.4	21.4	15.9	11.0		
		MCWB	8.9	8.0	10.6	13.9	17.4	20.4	20.2	19.8	18.3	14.9	11.7	8.6		
	2%	DB	10.1	11.1	16.5	21.3	24.6	29.3	30.1	29.7	24.4	19.4	13.5	9.5		
		MCWB	7.3	6.8	9.7	12.7	16.0	19.5	19.2	19.7	17.4	14.3	10.4	7.4		
	5%	DB	8.1	9.0	14.4	19.4	22.5	27.2	28.3	27.5	22.2	17.5	11.5	8.1		
		MCWB	5.9	5.9	8.7	11.9	14.9	18.7	19.0	19.1	16.3	13.7	9.3	6.4		
	10%	DB	6.2	7.0	12.1	17.1	20.3	25.0	26.2	25.2	20.2	15.8	9.9	6.4		
		MCWB	4.5	4.6	7.8	10.9	14.0	17.5	18.2	18.2	15.6	13.0	8.1	5.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	8.7	11.4	14.6	18.5	21.9	21.3	21.2	19.4	16.3	12.0	9.1		
		MCDB	11.6	12.6	16.9	22.2	25.1	29.8	29.9	29.1	24.9	19.6	14.9	10.6		
	2%	WB	7.7	7.2	10.2	13.2	17.1	20.3	20.2	20.2	18.0	15.1	10.7	7.8		
		MCDB	9.6	10.4	14.9	20.3	22.9	27.4	27.9	27.6	22.7	18.3	13.0	9.3		
	5%	WB	6.0	6.1	9.2	12.2	15.7	19.3	19.5	19.5	17.0	14.1	9.6	6.5		
		MCDB	8.0	8.8	13.5	18.4	20.7	25.7	26.6	26.1	21.1	16.8	11.3	7.9		
	10%	WB	4.6	4.7	8.1	11.2	14.6	18.3	18.7	18.7	16.1	13.2	8.3	5.1		
		MCDB	6.1	6.9	11.9	16.3	19.3	24.0	25.0	24.4	19.5	15.4	9.7	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	7.2	9.4	10.5	10.0	10.5	10.9	10.4	9.5	8.1	6.3	5.4		
		MCDBR	7.9	10.8	13.8	15.0	14.3	14.1	14.6	14.1	12.4	10.9	9.2	7.0		
	5% WB	MCWBR	5.6	7.0	7.9	7.7	6.9	6.2	5.7	5.5	6.4	6.6	6.3	5.3		
		MCWBR	7.3	10.2	12.3	13.7	12.4	13.0	13.0	12.6	11.0	9.8	8.5	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.316	0.354	0.392	0.404	0.406	0.406	0.397	0.374	0.350	0.315	0.288		
		taud	2.443	2.374	2.320	2.244	2.256	2.294	2.288	2.326	2.373	2.426	2.466	2.451		
	Ebn at Noon		807	851	865	861	861	859	854	847	834	800	766	764		
		Edh at Noon	67	88	108	128	131	127	127	117	102	83	66	59		
(44) All-Sky Solar Radiation	RadAvg		1.23	2.08	3.25	4.36	5.01	5.94	5.76	5.01	3.87	2.47	1.35	1.00		
	RadStd		0.15	0.31	0.48	0.64	0.46	0.55	0.50	0.46	0.35	0.24	0.17	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.47	N/A	N/A	N/A	N/A
(47) Regional (114 neighbors)		N/A	N/A	N/A	+0.75	+0.41	+0.31	N/A	N/A	+57	+16

Nomenclature: See separate page